宏观物体难以约束.

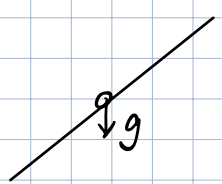
约束问题 { 平衡
稳定 / 不稳定

平衡：所有受力抵消.

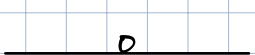
稳定：所有受力被阻尼

不稳定：所有受力被放大.

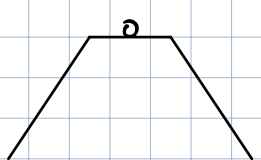
E.g.



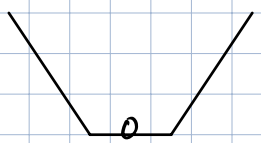
不平衡



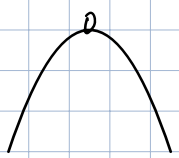
中性平衡



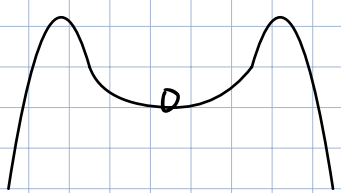
亚稳定平衡.

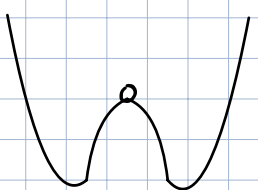


稳定平衡.



不稳定平衡





MHD 方程.

离子： $Mn\frac{\partial v_i}{\partial t} = en(\vec{E}+\vec{v}_i\times\vec{B}) - \nabla P_i + Mn\vec{g} + \vec{P}_{ie}$

电子： $mn\frac{\partial v_e}{\partial t} = -en(\vec{E}+\vec{v}_e\times\vec{B}) - \nabla P_e + mn\vec{g} + \vec{P}_{ei}$

两式求和：

$$n\frac{\partial}{\partial t}(M\vec{v}_i + m\vec{v}_e) = en(\vec{v}_i - \vec{v}_e)\times\vec{B} - \nabla\vec{P} + n(M+m)g$$

$\vec{v} = \frac{M\vec{v}_i + m\vec{v}_e}{M+m}$　　$\vec{j}$　　$\nabla(\vec{P}_i+\vec{P}_e)$　　ρ

$$\rho\frac{\partial\vec{v}}{\partial t} = \vec{j}\times\vec{B} - \nabla P + \rho g$$

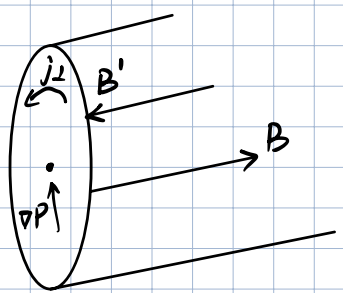
稳恒态下，$\frac{\partial}{\partial t} = 0$，$g = 0$.

$$\vec{j}\times\vec{B} = \nabla P$$

1. $\vec{B}\times(\vec{j}\times\vec{B}) = B^2\vec{j} - \vec{B}(\vec{j}\cdot\vec{B}) = B^2\vec{j} = \vec{B}\times\nabla P$

Def $\vec{j}_\perp\frac{\vec{B}\times\nabla P}{B^2} = (k_BT_i + k_BT_e)\frac{\vec{B}\times\nabla n}{B^2}$　　$P = nKT$

$\nabla P = KT\nabla n$



2. $\vec{j} \perp \vec{B} \perp \nabla P$. 磁力线，电流线都在一个等压面上.

$\nabla P = nKT$

3. $\nabla P = \vec{j} \times \vec{B}$. $\frac{\partial P}{\partial s} = 0$ 沿着磁力线，等离子体密度压强梯度不变.

s 为磁力线的弧长.

由

$$\begin{cases} \vec{j} \times \vec{B} = \nabla P \\ \nabla \times \vec{B} = \mu_0 \vec{j} \end{cases} \Rightarrow \vec{j} = \frac{1}{\mu_0} \nabla \times \vec{B} \Rightarrow \frac{1}{\mu_0} (\nabla \times \vec{B}) \times \vec{B} = \nabla P.$$

展开：

$$\frac{1}{\mu_0} (\nabla \times \vec{B}) \times \vec{B} = \frac{1}{\mu_0} [(\vec{B} \cdot \nabla) \vec{B} - \nabla \cdot \vec{B} \cdot \vec{B}]$$

$$= \frac{1}{\mu_0} [(\vec{B} \cdot \nabla) \vec{B} - \frac{1}{2} \nabla B^2]$$

假设 $\vec{B} \cdot \nabla \sim 0$.

$$= -\frac{1}{2\mu_0} \nabla B^2$$

$\therefore$ $\nabla (P + \frac{1}{2\mu_0} B^2) = 0$ or $P + \frac{1}{2\mu_0} B^2 = const.$

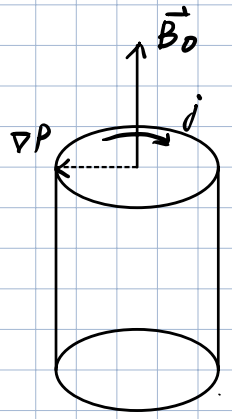
- 存在密度梯度的情形下，$P \uparrow B \downarrow$，$P \downarrow B \uparrow$.
- 强磁场压缩等离子体，
- 高密度处总磁场的减弱是由抗磁性电流产生.

Def 比压

$$\beta \equiv \frac{max(\sum n k_B T)}{max(\frac{B^2}{2\mu_0})}$$

一般，β 在 $10^{-6} \sim 10^{-3}$，抗磁性效应很弱. 因此往往可假设等离子体为均匀场.

when β 很大，抗磁性明显，高密度磁场被削弱.



$P_0 \cos^2 \frac{\pi r}{2a}$

$$\nabla P = \frac{\partial P}{\partial r} = \frac{\pi}{a} P_0 \cos\frac{\pi r}{2a} \sin\frac{\pi r}{2a} = \frac{\pi}{2a} \frac{B_0^2}{2\mu_0} \sin\frac{\pi r}{a}$$

1. $P_0 = \frac{B_0^2}{2\mu_0}$, $(\beta = 1)$.

2. $J \times B = \nabla P$.

$\nabla \times B = \mu_0 J$ $\quad$ $J \times B = \nabla P$.

$$\nabla \left(P + \frac{B^2}{2\mu}\right) = 0 = \nabla P + \frac{2B}{2\mu} \nabla B -$$

$$\frac{1}{\mu_0} (\nabla \times B) \times B = -\frac{1}{2\mu_0} \nabla B^2 = \nabla P.$$

$$\nabla B^2 = -2\mu_0 \nabla P = -\frac{\pi}{2a} B_0^2 \sin\frac{\pi r}{a}.$$

$$B^2 = \left(-\frac{B_0^2}{2} \cos\frac{\pi r}{a}\right)\Big|_0^r = \frac{B_0^2}{2} ($$